



A prediction model for the mechanical properties of SUS316 stainless steel ultrathin strip driven by multimodal data mixing

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ABSTRACT

Constructing a mapping relationship among material preparation process, microstructure, and mechanical properties is a challenge in material research and development. In this work, a deep learning framework for multimodal data fusion is constructed that couples a multi-layer perceptron (MLP) and a residual neural network (ResNet) to predict mechanical properties of SUS316 stainless steel ultrathin strips. Specifically, the MLP branch is used to extract the rolling process data features, and the ResNet with the addition of a convolutional block attention module (CBAM) is used to extract the microstructure features. Six models are constructed for comparison under the comprehensive consideration of factors such as unimodal network, the multimodal network and input form of image samples. The results show that the multimodal data model fused with the ResNet and MLP after adding the CBAM using both rolling process data and four types of microstructure image data as model inputs has the most accurate prediction results. The R^2 , MAPE, RMSE and MAE are 0.998, 0.727, 4.440 and 3.359, respectively. In addition, the proposed model is used for predicting yield strength and elongation, and the results show that the R^2 values of both models on the test set are greater than 0.980, fully confirming that the multimodal data model has high prediction accuracy and good generalizability.

1. Introduction

Typically, stainless steel strips with thicknesses between 0.01 mm and 0.1 mm are referred to as precision stainless steel ultrathin strips and are typically prepared through rolling techniques. Precision stainless steel ultrathin strips are widely used in electronic communication, medical equipment, and instrumentation because of their significant advantages of high precision, light weight, and high performance. Mechanical properties such as tensile strength, yield strength, and elongation are important indicators of the quality of precision stainless steel ultrathin strip products and can also be used to determine whether the final performance of downstream products can meet the requirements for use. Therefore, accurately predicting the mechanical properties of ultrathin stainless steel strips is important for guiding rolling process optimization and product performance control.

The existing modeling methods for predicting the mechanical performance of plates and strip steels can be roughly divided into two types: mechanism models, which are based on the experimental results of laboratory physics and metallurgy, and models that predict the

mechanical properties of products, such as mathematical models used to predict the organizational evolution and mechanical properties of steel materials. For example, magnesium aluminum alloy sheets were prepared via different rolling processes, and then, on the basis of the Hall-Petch relationship, nonlinear fitting was performed on the tensile strength and average grain size while considering the internal grain size distribution of the sheet structure. An accurate model for predicting the tensile strength of magnesium alloy sheets after rolling was constructed, achieving high-precision prediction in predicting related properties [1]. In addition, a new high-temperature creep constitutive model coupling dislocations, phase structure, and damage was established via the phase field method framework, which combines theories such as thermodynamics, elastoplastic mechanics, and damage mechanics to predict the creep behavior of nickel based single crystal high-temperature alloys in high-temperature environments [2]. Although the study of mechanism models has received widespread attention, such methods have certain limitations. Specifically, the model structure is complex, involving multiple physical field couplings and microstructure descriptions, with high difficulty and high computational complexity. A large number of

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precise parameters are needed, and obtaining precise parameters through experiments is costly and difficult. Poor adaptability to new materials and complex working conditions are also encountered. Model assumptions and parameter uncertainties can lead to biased prediction results.

The other type is data-driven modeling, which is based on a large amount of data, and uses artificial intelligence, machine learning and data mining technology to predict material performance faster, more accurately and more economically [3]. Early data-driven modeling research mainly used artificial neural networks to predict the mechanical properties of various metals after forming [4,5], including establishing prediction models for the mechanical properties of welded seams and seamless steel pipes after rolling on the basis of artificial neural networks [6,7]. This type of model has a simple structure, and the adjustment of neural network hyperparameters relies mainly on manual testing. The development of heuristic intelligent optimization algorithms provides an effective way to optimize of hyperparameters in neural networks. Therefore, by combining production data, artificial neural networks, and intelligent optimization algorithms, more accurate material mechanics performance prediction models can be constructed [8], which provides a foundation for the online application of the models. In recent years, traditional artificial neural networks have been unable to fully meet the needs of metal material performance prediction modeling, and various intelligent methods and deep learning methods have also been used in related research [9–11]. Dai et al. [12] used deep learning to overcome the complexity of the physical modeling process of the mechanical properties of high-entropy alloys, and the predicted room-temperature properties were in good agreement with experimental measurements. Xie et al. [13] formulated an optimal deep neural network based on process parameters and steel metal composition parameters for online monitoring and control of the mechanical properties of steel and guided the production of target steel plates with customized mechanical properties. Yang et al. [14] built a time series neural network based on long short-term memory (LSTM) for predicting the yield strength, ultimate tensile strength, and elongation of steel plates via the production process parameters of the steel and important information related to the time series. The results verify that the proposed LSTM model has sufficient accuracy and outperforms conventional algorithms (SVM, RF, and ANNs). The common feature of the above studies is the combination of preparation process parameters and machine learning methods to achieve accurate prediction of mechanical properties. However, providing microstructural data for training in numerical form rather than raw images may result in missing key information.

In addition to building a model that maps the relationships between process parameters and mechanical properties, establishing a mechanical property prediction model for the correspondence between the microstructure and macroscopic properties of materials is also a hot topic in scientific research. Xu et al. [15] combined the microstructure characteristics of composite materials with transfer learning algorithms to construct an accurate model for predicting the mechanical properties of composite materials, and verified the high efficiency and accuracy of the method. Guo et al. [16] used discrete element simulation and Potts dynamic Monte Carlo simulation to synthesize the microstructure of composite materials and used it as input data for the CNN model. The corresponding macroscopic properties were used as labels for training, and a mechanical performance prediction model with better generalizability and robustness under the conditions of data augmentation treatment technology was constructed. Ye et al. [17] proposed a machine learning-based approach to predict the effective elastic properties of composites with arbitrary shapes and inclusions distribution, demonstrating that the trained network can effectively establish an accurate mapping relationship between the effective mechanical properties and the microstructures of complex structural composites. Xu et al. [18] converted production data into two-dimensional data images and predicted the mechanical properties of hot rolled strips via chemical

composition and process parameters based on the CNN method, and the results were consistent with measurements of the performance of the steel used in the experiment. Wang et al. [19] comprehensively considered the laser powder bed melting process parameters and microstructure, extracted the relevant microstructural features, and constructed a more accurate tensile strength prediction model for Ti6Al4V via machine learning. Yan et al. [20] investigated the relationships among microstructures such as secondary dendrite arm spacing (SDAS), grain size, eutectic silicon morphology, and tensile mechanical properties of cast Al7Si1.5Cu0.4Mg alloys at different temperatures. On the basis of the relationship between microstructures and hardness, a model for predicting the ultimate tensile strength was proposed and achieved good prediction results. Xu et al. [21] established a model for predicting the mechanical properties of carbon nanotube/aluminum (CNT/Al) composites on the basis of the microstructure of the composites, and the model predictions matched the results of tensile experiments. Fu et al. [22] developed a model mapping the relationship between the microstructure and mechanical properties of ZE41-xCa-ySr alloys ($0 \leq x \leq 2$, $0 \leq y \leq 0.2$) via an ANN and multiple regression, which yielded an optimum aging temperature of 322°C and a corresponding aging time of 11h for ZE41-0.17Ca-0.2Sr alloys. Abueidda et al. [23] developed a CNN model to quantitatively predict the mechanical properties of 2D tessellated composites and combined it with a genetic algorithm optimizer to determine the optimal microstructure. Ren et al. [24] used a deep learning (DL) strategy using microstructural images as direct inputs to predict the tensile properties of dual-phase steel, which overcomes the difficulty of quantifying complex microstructural information and had high portability and applicability. He et al. [25] established a reliable model for predicting the tensile strength of steel on the basis of alloying elements, metallographic organization, and production process parameters, and the results revealed that the proposed radial basis function neural network model has good prediction accuracy. Wang et al. [26] formed a modeling dataset by deeply integrating the mechanical properties and microstructure evolution mechanism of ultrathin niobium strips. On this basis, a data-driven intelligent model for predicting mechanical properties was established via the XGBoost ensemble learning algorithm. Machine learning algorithms can be used to build a model that maps the relationship between process parameters and mechanical properties during material processing or a model that maps the relationship between microstructure and mechanical properties, in both cases, the input of the model is a single data type or data format. Single modal data-driven modeling may lead to problems such as the incomplete mapping of relationships and difficulty in meeting the requirements of prediction accuracy [27,28].

With the development of deep learning, multimodal data fusion-driven modeling has become a recent research hotspot and has been applied in many fields. Multimodal data fusion-driven modeling is the process by which a model processes different forms of data (e.g., image, video, text, and speech) in analysis and recognition-related tasks, which can provide more information for the model's decision-making. Li et al. [29] used textual (macroscopic properties of composites and matrices) and visual (two-point statistics of microstructure) data to predict the material properties of carbon fiber monofilaments. The developed hybrid CNN-MLP model achieved highly accurate predictions of the longitudinal modulus, transverse modulus, in-plane shear modulus, principal Poisson's ratio, and out-of-plane shear modulus of carbon fibers. Song et al. [30] proposed a multimodal data fusion prediction model fusing material microstructure information and material composition information, which was validated on data from alternative composites, providing an effective solution for the measurement and prediction of material properties. Ren et al. [31] developed a deep learning framework for processing multimodal databases, including compositional and multisource microstructural images, for predicting the tensile properties of dual-phase steels. The framework provides useful guidance for the prediction of composition-microstructure-property relationships in complex steel systems and can be generalized

to other alloys. Zhao et al. [32] designed a neural network-based microstructure identification model and a performance prediction model for titanium alloys. Through the fusion of the two models, a heterogeneous data model of the titanium alloy composition (data), microstructure (images), and properties (curves) was established, and this work confirms that multimodal data fusion can effectively solve the problem of the complexity of the relationships among different forms of material data.

Currently, the mechanical performance prediction models based on multimodal data are mainly models that jointly input material composition and microstructure images. In the actual rolling production process, the rolling process parameters setting directly affects the generation and evolution of the internal microstructure of the strip, thereby determining the final mechanical properties of the strip. Although the cost of obtaining the mechanical properties of a strip is generally lower than that of obtaining microstructure images, this method often involves post inspection. If a multimodal data-driven mechanical property prediction model is constructed using powerful deep learning methods and accumulated historical data in the laboratory, including two modal data of microstructure images and process data, the constructed model can be combined with the software for microstructure evolution analysis of metal processing to achieve metal material preparation process design and optimization. The specific ideas are as follows: In order to obtain the rolling preparation process of stainless steel ultrathin strips with specific target mechanical properties, the rolling process can be designed first, and the designed process parameters can be input into the software of microstructure evolution analysis in metal processing to simulate the microstructure evolution image data of the strip under the process, including the size, shape, distribution of grains, and the type, proportion, and degree of recrystallization of the phases. Secondly, the pre-designed rolling process and the microstructure image data obtained by simulation are jointly input into the established multimodal data-driven mechanical properties prediction model for mechanical properties prediction. Then, the predicted mechanical properties are compared with the target mechanical properties. If the gap is large, the original design rolling process is revised and the above operation is repeated until the target mechanical properties are achieved. Finally, according to the iterative optimization rolling process, the actual rolling preparation of stainless steel ultrathin strip is carried out under laboratory conditions and the optimized rolling process and predicted mechanical properties are verified to meet the actual requirements. The whole process can realize the design and optimization of stainless steel ultrathin strip rolling process under the premise of reducing the number of real experiments and the cost of trial and error, and improve the efficiency of process development. Therefore, it is necessary to establish a mechanical performance prediction model for stainless steel ultrathin strips driven by the coupling of process parameters and microstructure image data. This research can not only promote the efficiency of ultrathin strip rolling process development but also guide the development of new products with specific excellent mechanical properties, providing support for the innovative design of ultrathin strip preparation.

This paper presents a deep learning framework based on multimodal data fusion to predict the tensile strength of SUS316 stainless steel ultrathin strips. The rolling process data and image data spliced with the EBSD inverse pole figure, recrystallized fraction, local average misorientation, and Schmidt's factor map are used as joint inputs for model training, and the CBAM mechanism is introduced to increase the ability of the CNN to extract the image features of the material microstructure. The model not only considers the role of the rolling process on the mechanical properties of the product but also considers the influence of the microstructure, which further improves the prediction accuracy and generalization performance of the model.

The rest of the paper is organized as follows. The process of acquiring the dataset used for modeling and the preprocessing of the experimental data are presented in Section 2. The framework of the method and the

Table 1

SUS316 stainless steel rolling process.

Passes	Inlet thickness/ μm	Exit thickness/ μm	Front tension/ kN	Back tension/ kN
1	55	50	1.94	2.32
2	50	40	1.80	2.00
3	40	35	1.68	1.80
4	35	30	1.50	1.70
5	30	28	1.42	1.58

multimodal fusion model for predicting the mechanical properties of ultrathin strips of stainless steel are presented in Section 3. The experiments and results of the method are discussed in Section 4. Finally, the paper is summarized in Section 5.

2. Construction of the modeling dataset

2.1. Acquisition of rolling process data

SUS316 stainless steel strips with an initial thickness of 55 μm are selected as the raw material for the rolling experiments, and the width of each strip is 150 mm. Comprehensive consideration of the deformation of each pass, the maximum rolling force limit of the mill, and the yield strength of the material to design the rolling process, as shown in Table 1. The experiment was carried out with a 280 mm twenty-roll mill of the Taiyuan University of Technology, with a work roll material of M2 (grade W6Mo5Cr4V2), work roll diameter of Φ16 mm, roll body length of 280 mm, and maximum rolling force of 400 kN, and the roll system is lubricated by rolling oil in the spray state between the rolls. The rolling mill used is shown in Fig. 1.

The rolling method adopts multipass reversible rolling, the maximum rolling speed is 30 m/min, the front and back tension in the rolling process is measured by a tensiometer, and the inlet thickness and outlet thickness of each pass are measured by a thickness gauge. The schematic diagram of the rolling process of SUS316 stainless steel ultrathin strip is shown in Fig. 2. An ultrathin strip of SUS316 stainless steel in the rolled state was used as the object of study, and heat treatment was carried out in a CHY-1700 vacuum tubular heating furnace, with annealing temperatures of 800, 850, 900, 950, 1000, and 1050 °C. The tube was heated to 300 °C when the argon gas and ultrathin strips entered the furnace, and the furnace was heated to at the set temperature for 15min while the furnace cooled to 300 °C.

Fig. 3 shows the global mapping of the correlation of the rolling process data (thickness (TK), number of rolling passes (RP), annealing temperature (AT), depression rate (DR), cumulative depression rate (CDR), rolling speed (RS), rolling force (RF), back tension (IT), and front tension (OT)), and the matrix shows the degree of correlation between the process parameters.

Preprocessing of the data is required before the initial data are entered into the model, as the data vary greatly from one data point to another in terms of quantities and units. To eliminate this discrepancy, the process data were normalized via the sklearn tool according to the min-max normalization method so that each process data point transformed value maps between [0–1]. The transformation formula is as follows:

$$x^* = (x - x_{\min}) / (x_{\max} - x_{\min}) \quad (1)$$

where $x_{\max}$ is the maximum value of the sample data and $x_{\min}$ is the minimum value of the sample data.

In this study, the heat treatment temperature was divided into six gradients, the heat treatment temperature gradients were processed with one-hot coding to convert the heat treatment temperature into one-hot vectors, and the converted one-hot vectors are shown in Table 2.

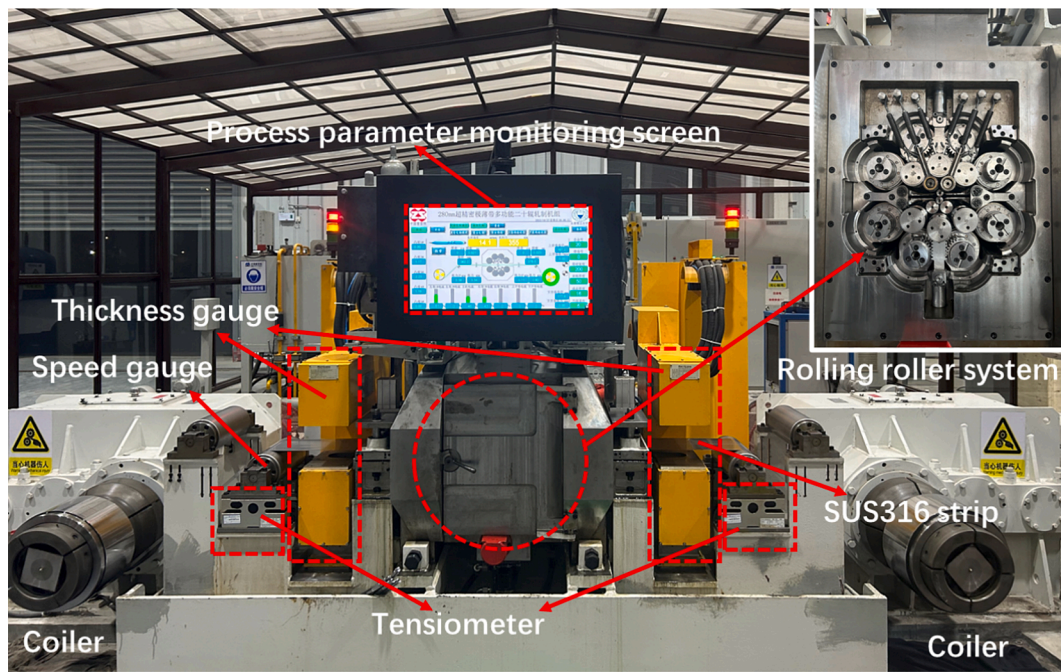


Fig. 1. Precision stainless steel ultrathin strip rolling platform.

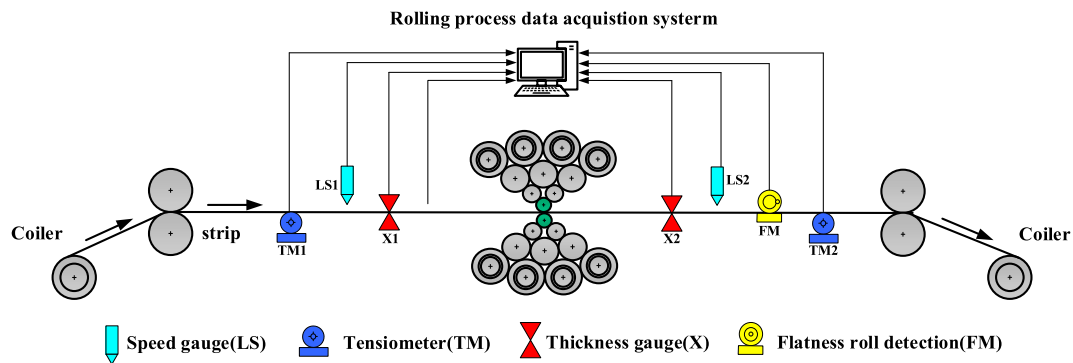


Fig. 2. Flowchart of the SUS316 stainless steel ultrathin strip rolling process.

2.2. Acquisition of mechanical property data

Annealed stainless steel ultrathin strips are made into tensile samples. Unidirectional tensile test is conducted on INSTRON5969 electronic universal tensile testing machine. The test temperature is room temperature, the tensile rate is 1 mm/min, and the number of tensile tests performed on the sample at different passes and different temperatures is 5 with obvious outliers or error values rejected, and valid data retained. Fig. 4. shows the tensile strength, yield strength and elongation test results for the stainless steel strips in different states.

2.3. Acquisition of microstructure images

The inverse pole figure (IPF), recrystallized fraction (RF), local average misorientation (LAM), and Schmidt factor map (SF) are obtained via EBSD characterization, and the resolution of each image is 650×511 . If the 650×511 EBSD image is scaled directly to 128×128 , the grain size and shape information in the EBSD image changes, which affects the mechanical properties. Therefore, it is necessary to ensure that the input image contains as many polycrystalline grains as possible without the loss of detailed grain features. The 650×511 EBSD image is cropped to four 325×255 images. The cropped image is redefined to

have a size of 128×128 , and the four types of EBSD images of the same size are subsequently stitched together into an image with an overall size of 256×256 pixels. A schematic diagram of image processing and splicing is shown in Fig. 5.

The training of deep learning models depends more on the training data. To further improve the diversity of microscopy image data in the training dataset, image enhancement [33–37] is used to increase the number of microstructure images to improve the stability of the prediction model. By using vertical flipping, horizontal flipping, clockwise rotation of 90 degrees, clockwise rotation of 180 degrees, and counter-clockwise rotation of 90 degrees, each image is cropped and enhanced to yield 24 images, resulting in a total of 864 sets of images.

3. Theory, methodology, and model

3.1. Background and framework

Many researchers have established mechanical property prediction models based solely on rolling process data or microstructure images but have not fully considered the comprehensive impact of both on mechanical properties, which may lead to incomplete mapping relationships. This work adopts a multimodal data fusion mechanical

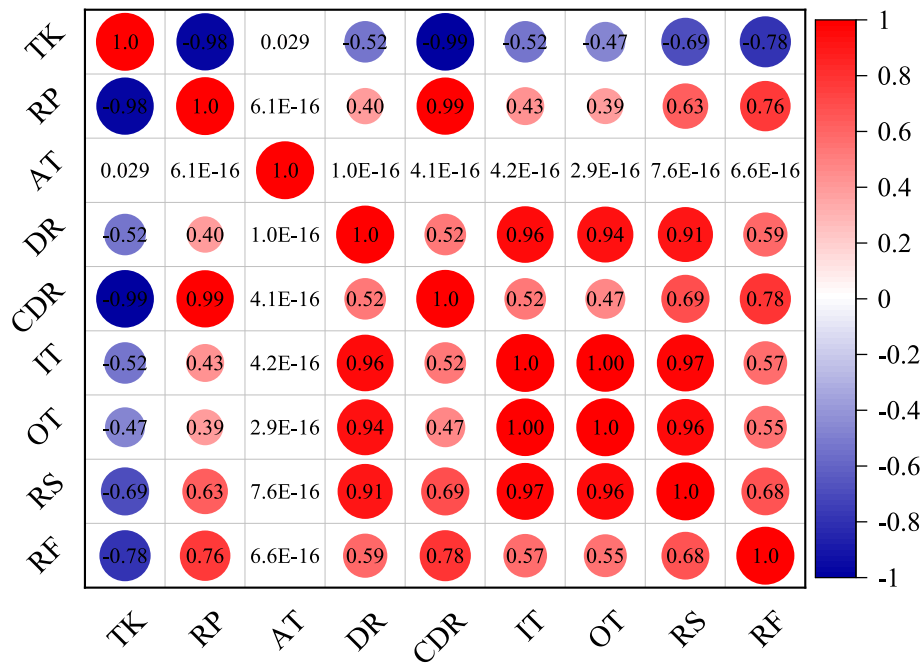


Fig. 3. Characteristic correlation coefficients.

Table 2

One-hot coding processing results for the heat treatment temperatures.

	800°C	850°C	900°C	950°C	1000°C	1050°C
0	1	0	0	0	0	0
1	0	1	0	0	0	0
2	0	0	1	0	0	0
3	0	0	0	1	0	0
4	0	0	0	0	1	0
5	0	0	0	0	0	1

performance prediction model that integrates the rolling process and microstructure data to construct a model with more comprehensive mapping relationships and higher prediction accuracy.

The architecture of multimodal fusion includes a joint architecture and a collaborative architecture [38,39]. The joint architecture maps data from different modalities into a shared semantic subspace to fuse multimodal features. The goal of the cooperative architecture is to find relationships between different modal data. The main objective of this study is to improve the prediction accuracy of mechanical properties on the basis of the complementarity of information between the microstructure and rolling process of ultrathin stainless steel strips. Therefore, the main multimodal fusion framework used in this study is the joint architecture, which mainly extracts high-level features from the input data of different modalities through multiple networks and then fuses the high-level semantic features.

The microstructure of ultrathin stainless steel strips can be characterized by EBSD, and CNNs have excellent image feature extraction capabilities for capturing features such as grain size, grain boundaries, recrystallized grains, subcrystallized grains, and deformed grains. Therefore, a CNN is used to extract high-level semantic features of the microstructure of ultrathin strips of stainless steel. However, not all microstructures in the EBSD images are correlated with mechanical properties, so the model needs to focus more on important regions when extracting microstructural features. Therefore, the CBAM is added to the CNN so that the model pays more attention to the important microstructures in the microimages that are highly relevant to the output result during prediction.

The rolling process of a stainless steel ultrathin strip is characterized

by an MLP to extract features, and the actual rolling process information includes not only continuous variables such as the rolling force and rolling speed but also discrete variables such as the annealing temperature. Inputting two types of variables into the MLP to extract high-level features is obviously unreasonable and makes it difficult to learn the underlying features interactively. Therefore, by adding the one-hot coding module [40,41], discrete variables are changed into a binary format, which facilitates the input of category-type data as features into the multilayer perceptual machine, and to a certain extent, it also expands the features. On the basis of the above proposed framework and discussion, we design a multimodal mechanical performance prediction model with the architecture shown in Fig. 6.

3.2. Multimodal fusion model for predicting mechanical performance

The proposed multi-mode fusion model for predicting the mechanical performance of stainless steel ultrathin thin strips is shown in Fig. 7 and includes the microstructure feature extraction network for extremely thin stainless steel strips, the rolling process feature extraction network, the feature vector fusion module and the fully connected layer for regression prediction.

3.2.1. Microstructure feature extraction network

ResNet is used to extract microstructure features. ResNet is not constructed by directly stacking convolutional layers but rather uses residual connectivity structures, successfully solving the problems of model degradation and training difficulties [42]. The material microstructure feature extraction network is further improved on the basis of ResNet18. The original network contains five stages, each with two Res-blocks. Since ResNet18 is designed for ImageNet, a large-scale dataset with multiple categories, the depth and width of the model are relatively large. On small-scale datasets, this can lead to slow training and overfitting. Therefore, in this study, the network structure is adjusted to adapt it for material microstructure property extraction by removing the original fifth stage and setting the number of channels for the first four stages to 16, 16, 32, and 64. The features of each channel are fused via a 1×1 convolutional layer to obtain 32 features that can reflect the material microstructure information, and the material microstructure feature vectors are obtained via global average pooling (GAP).

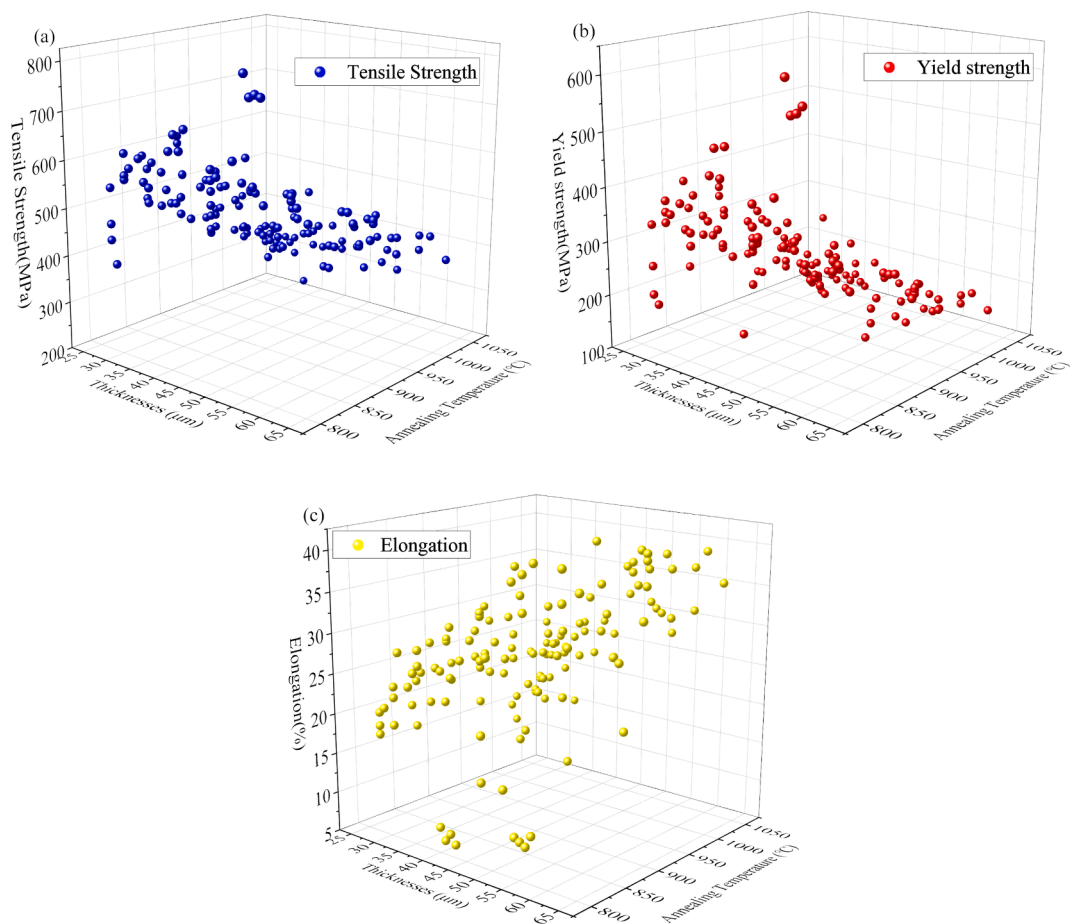


Fig. 4. Distribution diagram of mechanical property test samples for 316 stainless steel ultrathin strip. (a) tensile strength; (b) yield strength; (c) elongation.

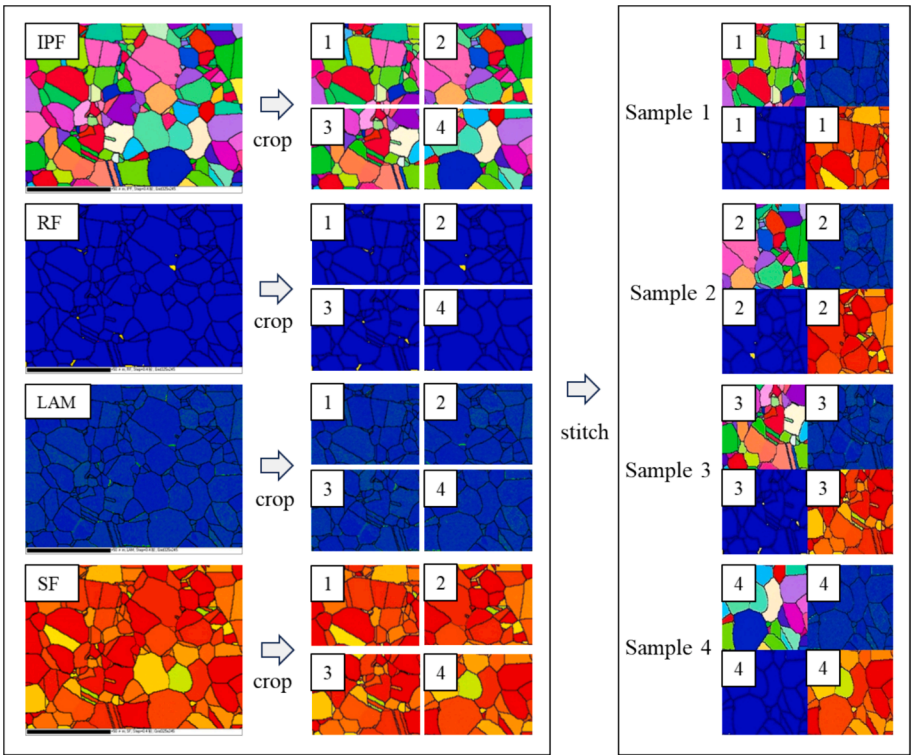


Fig. 5. Schematic diagram of image processing and splicing.

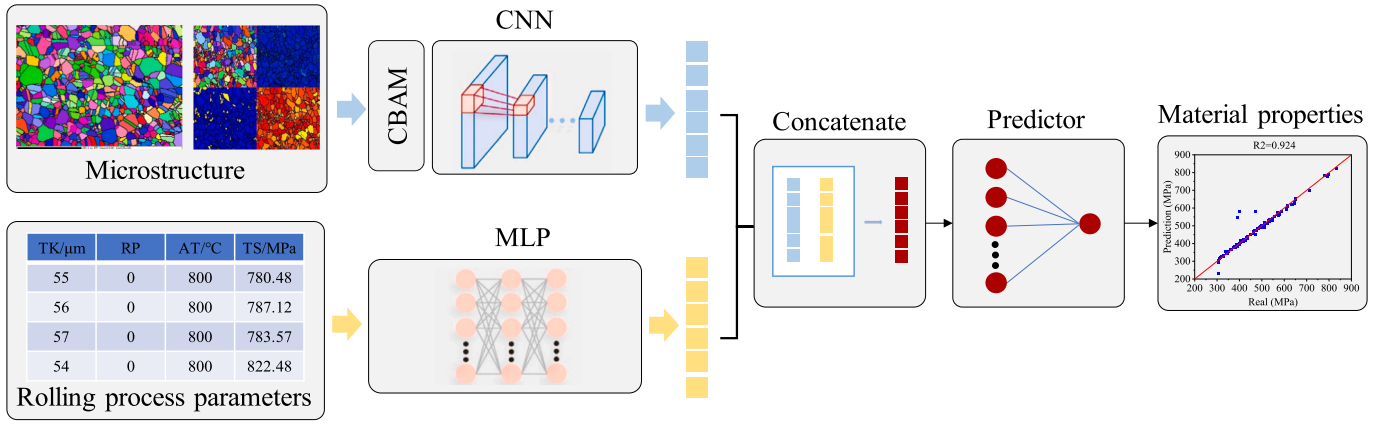


Fig. 6. Multimodal fusion model framework for predicting the mechanical properties of ultrathin strips of stainless steel.

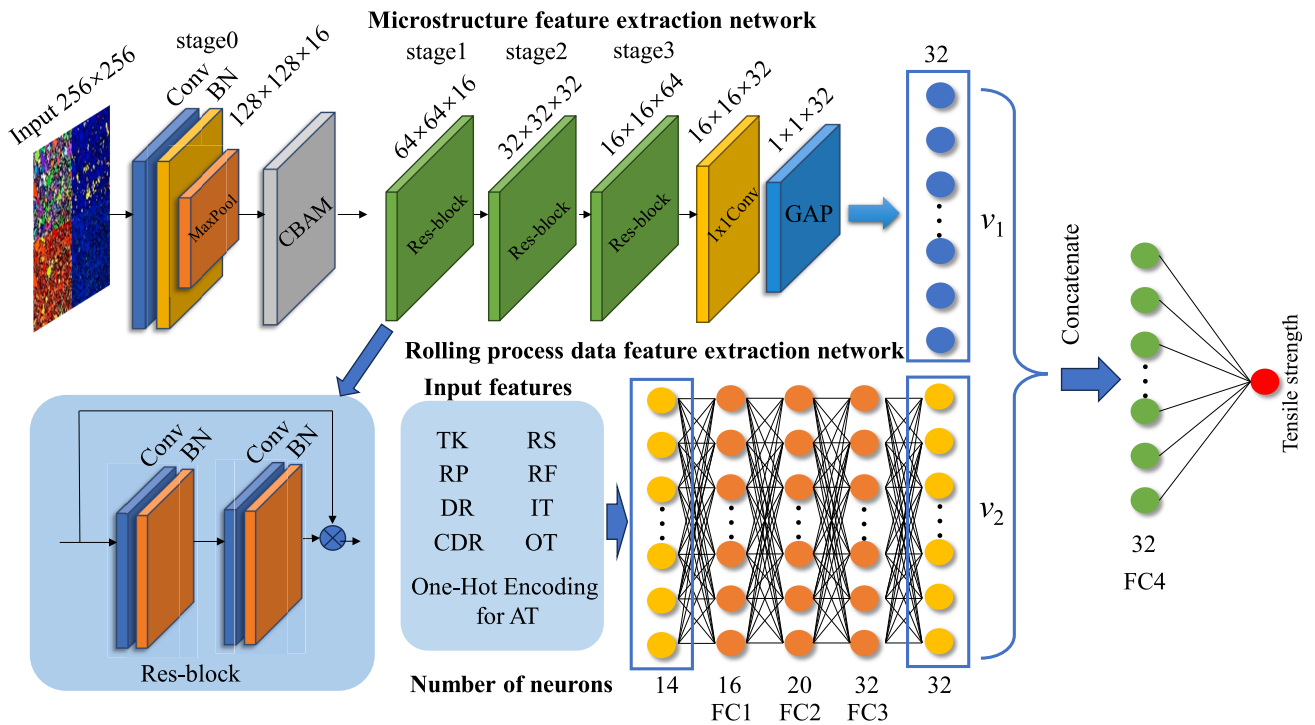


Fig. 7. Structure of the model for predicting the mechanical properties of ultrathin thin strips with multimodal data fusion.

The size of the resulting FeatureMap is assumed to be $h \times w \times c$, where h and w are the height and width of a single feature map, respectively, and c is the number of channels (the number of feature maps). Let v_1^k be the k th element GAP of the processed material microstructure feature vector v_1^k , and let $FeatureMap_{ij}^k$ denote the value of the k th feature map in the i th row and j th column; then, the GAP function can be expressed as follows:

$$v_1^k = \frac{1}{h \times w} \sum_{i=1}^h \sum_{j=1}^w FeatureMap_{ij}^k \quad (2)$$

ResNet18 consists of multiple convolutional layers with different numbers of convolutional kernels. Convolution operations with different depths can extract features at different levels, whereas multiple convolution operations with different widths can extract multiple types of features at the same level. Fig. 8 shows the EBSD image features extracted by ResNet18, which were trained on the multimodal dataset. The feature maps for each of the 5 stages are visualized as shown in

Fig. 8. The information reflected by the various feature maps extracted from the same stage is inconsistent. Taking the feature maps extracted from Stage 0 as an example, the 16 feature maps reflect many different types of features in the EBSD images, such as the grain size and degree of recrystallization, which have different effects on the mechanical properties of ultrathin strips of stainless steel [43]. Additionally, the presence of grains in the feature map of the same stage is accompanied by invalid regions that do not contribute to the prediction of the mechanical properties of the material.

The microstructure image contains not only microstructural information of various metallographic organizations in ultrathin strips of stainless steel but also irrelevant information such as voids and defects. To make the model pay more attention to the important regions, the CBAM is added, and the structure of the CBAM is shown in Fig. 9. The CBAM includes the CAM and SAM, the CAM can be used to increase the weight of feature maps that have an impact on the final task, and the SAM increases the weight of regions in the feature maps that are highly relevant to the task [42,44,45], e.g., allowing the model to add more

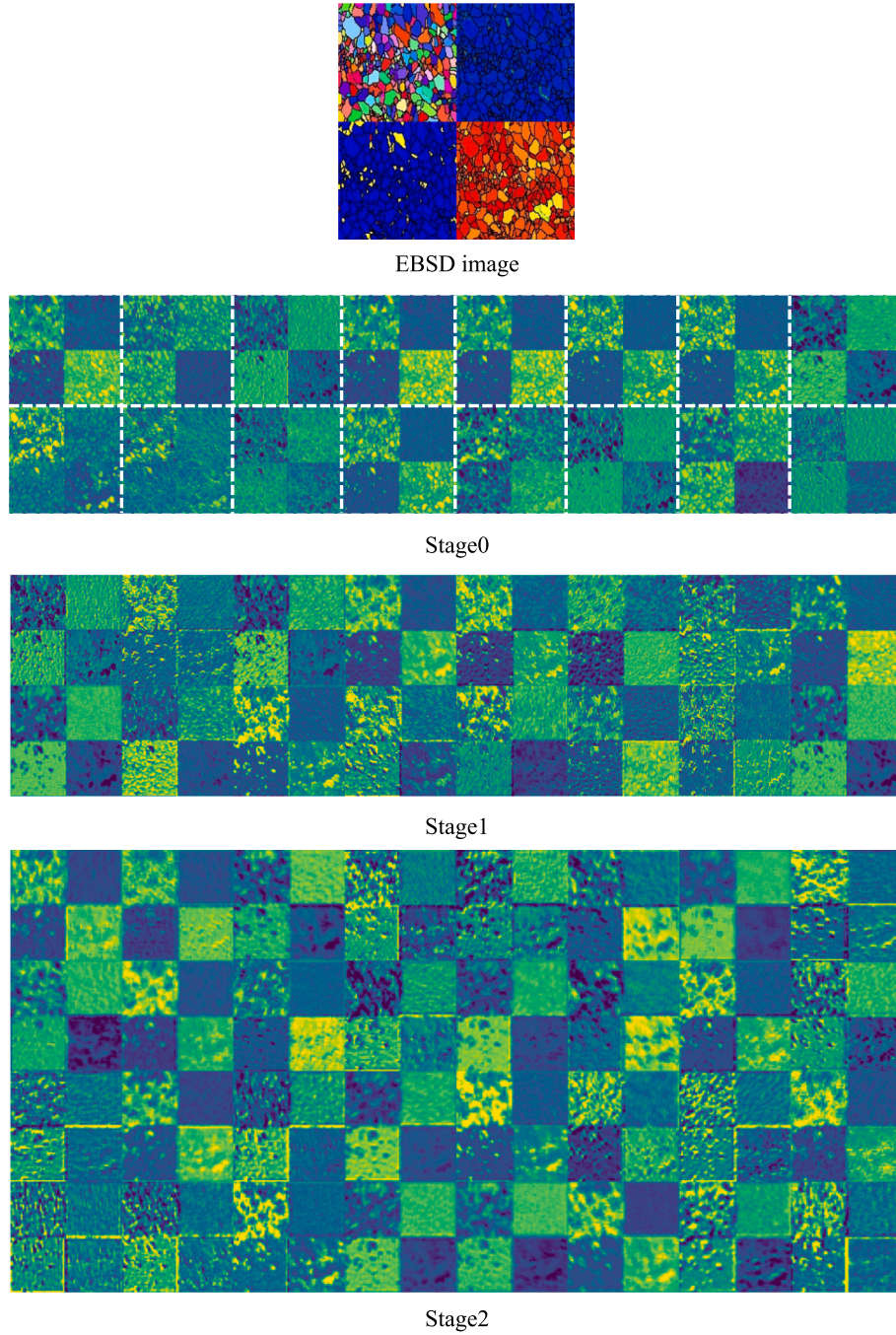


Fig. 8. Feature maps extracted from different depths of convolutional layers of different depths.

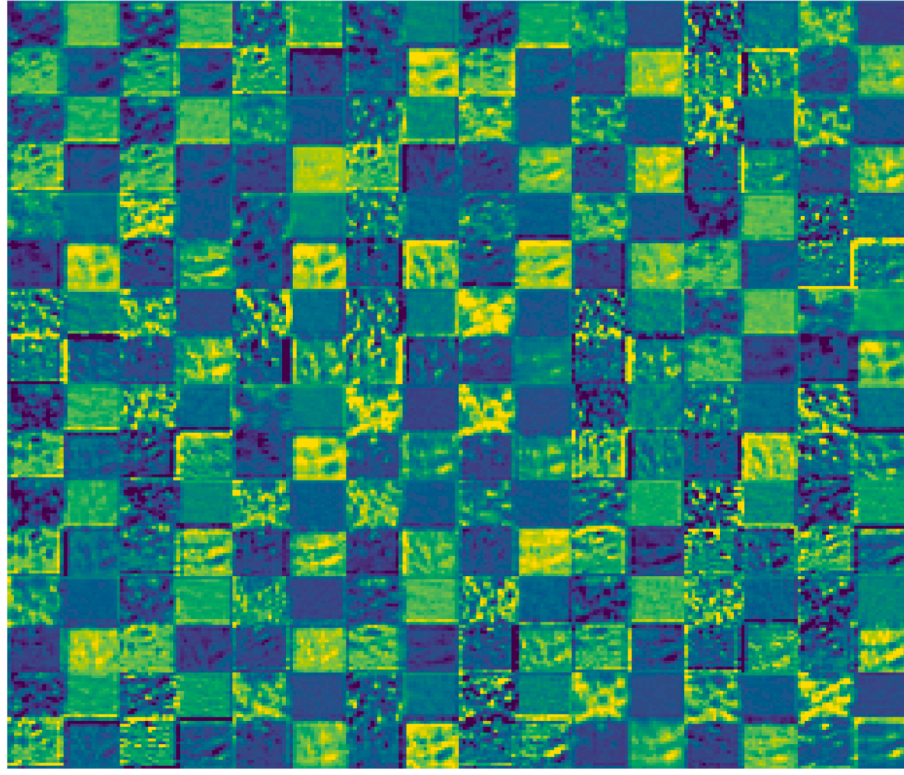
weight to focus on regions relevant to the mechanical properties rather than gaps. By adding the CBAM, the model learns which feature maps are more important and which regions on the feature maps are more important, thus improving the quality of the extracted microstructural features and increasing the accuracy of the model in predicting the mechanical properties of ultrathin strips of stainless steel.

Let the input features of the CAM be $F \in \mathbb{R}^{h \times w \times c}$, and let the features after the average pooling and maximum pooling operations be $F_{C-avg} \in \mathbb{R}^{1 \times 1 \times c}$ and $F_{C-max} \in \mathbb{R}^{1 \times 1 \times c}$. It then passes through a shared MLP with two fully connected layers. The weights of the two fully connected layers are denoted W_0 and W_1 . The channel attention weight $W_C \in \mathbb{R}^{1 \times 1 \times c}$ obtained by the CAM and the feature $F_O \in \mathbb{R}^{h \times w \times c}$ obtained by the CAM can be expressed as the following equation [30]:

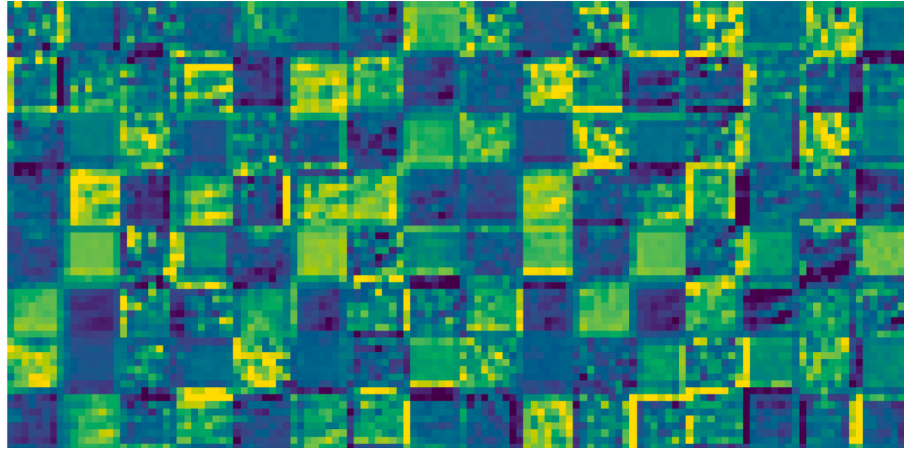
$$W_C = \text{Sigmoid}(\text{MLP}(\text{AvgPool}(F))) + \text{MLP}(\text{MaxPool}(F)) \\ = \text{Sigmoid}(W_1(W_0(F_{C-avg}))) + W_1(W_0(F_{C-max})) \quad (3)$$

$$F_O = F \otimes W_C \quad (4)$$

Let the input features of the SAM be $F' \in \mathbb{R}^{h \times w \times c}$, and let the features after the average pooling and maximum pooling operations be $F'_{S-avg} \in \mathbb{R}^{h \times w \times 1}$ and $F'_{S-max} \in \mathbb{R}^{h \times w \times 1}$, respectively. They are then connected and convolved through a convolutional layer. $\text{Conv}^{7 \times 7}$ denotes a convolution operation with a filter size of 7×7 . The spatial attention weight $W_S \in \mathbb{R}^{h \times w \times 1}$ computed by the SAM and the feature $F'_O \in \mathbb{R}^{h \times w \times c}$ obtained by the SAM can be expressed as the following equation [30]:



Stage3



GAP

Fig. 8. (continued).

$$\begin{aligned} \mathbf{W}_s &= \text{Sigmoid}(\text{Conv}^{7 \times 7}([\text{AvgPool}(\mathbf{F}); \text{MaxPool}(\mathbf{F})])) \\ &= \text{Sigmoid}(\text{Conv}^{7 \times 7}([\mathbf{F}_{s-\text{avg}}; \mathbf{F}_{s-\text{max}}])) \end{aligned} \quad (5)$$

$$\mathbf{F}'_o = \mathbf{F} \otimes \mathbf{W}_s \quad (6)$$

The structural parameters of the material microstructure feature extraction network are summarized in Table 3.

3.2.2. Feature extraction network for the rolling process parameters

The feature extraction network for the strip rolling process consists of an MLP. The input to the MLP network is the rolling process data, which then pass through three fully connected layers with 16, 20, and 32 neurons per layer. The ReLU activation function is added after the first two fully connected layers to increase the feature extraction capability of the rolling process data, and the third fully connected layer is used to output the rolling process data feature vector. Importantly, in this work,

the annealing temperatures are coded as category-type data with one-hot categories, which are also input as vectors into this feature extraction network. This network is shown in Table 4.

3.2.3. Regression predictor

A fully connected MLP layer containing 32 neurons is used as a regression predictor whose main function is to output the tensile strength of the material on the basis of the concatenate layer containing microstructure information and rolling process information.

Assume that $\mathbf{x}_1$ and $\mathbf{x}_2$ represent microstructure, images and rolling process data, respectively; g_1 and g_2 represent the microstructure feature extraction network and rolling process feature extraction network, respectively; f denotes the regression predictor; and Y represents for the mechanical properties of stainless steel strips [46,47], such as tensile strength. The process of predicting the mechanical properties can be expressed as the following equation:

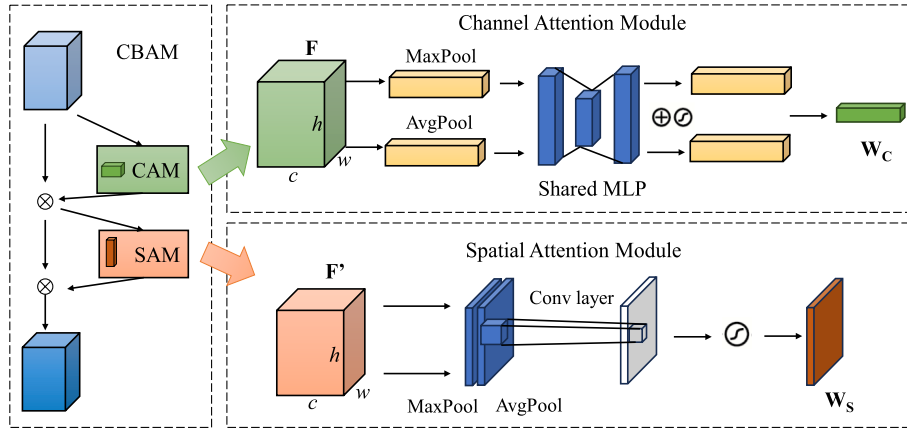


Fig. 9. Convolutional attention module.

Table 3
Detailed information of the microstructure feature extraction network.

Name	Output	Structure
Stage0	128×128×16	7×7, 16, stride of 23×3max-pooling, stride of 2
Attention	128×128×16	CBAM
Stage1	64×64×16	$\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 2$
Stage2	32×32×32	$\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 2$
Stage3	16×16×64	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
Compression	16×16×32	$[1 \times 1, 32] \times 1$
GAP	1×1×32	GAP
Output	1×1×32	v_1

Table 4
Detailed information of the feature extraction network for the rolling process data.

Name	Activation	Number of neurons
Rolling process feature	/	14
FC1	ReLU	16
FC2	ReLU	20
FC3	ReLU	32
Rolling process vector v_2	/	32

$$Y = f(g_1(x_1), g_2(x_2)) \quad (7)$$

Fig. 10 compares the prediction results of the concatenate layer splicing neuron method and the multimodal coupling method in this study. For the spliced neuron method, the model accuracy depends on the number of neurons used in the rolling process data and microstructural images, and the MAEs are 5.45%, 4.78%, 3.23%, 5.40%, and 14.00% when the ratios of the number of neurons used in the image and the rolling process data are 7:1, 3:1, 1:1, 1:3, and 1:7, respectively. The model accuracy increases with increasing proportion of the number of neurons in the rolling process, and when the proportion is 1:1, the MAE reaches the lowest value and then increasing again, which indicates that the rolling process information is as important as the microstructure information.

For the key parameters of the regression predictor, the number of neurons in the FC4 layer determines the main architecture of the regression predictor and should be optimized to improve the prediction accuracy. Therefore, this work systematically analyzes the effect of the number of neurons in the FC4 layer on the model performance to obtain the best prediction accuracy. The comparison results are shown in Fig. 11, where the MAE of the model decreases and then increases as the

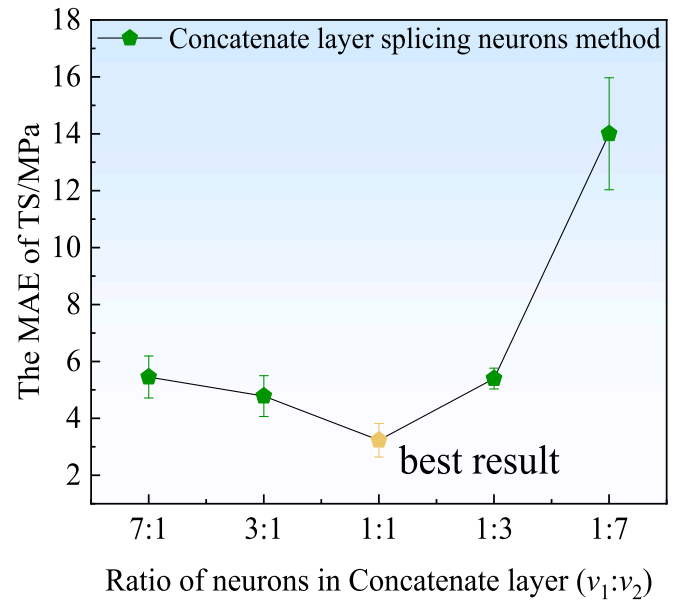


Fig. 10. Concatenate layer comparison of the scale of the image data: rolling process data.

number of neurons increases from 8 to 128. Apparently, 32 neurons is the optimal number of neurons, proving that the current selected architecture is suitable for the database in this paper and can accurately predict the tensile strength of ultrathin thin strips of stainless steel.

3.2.4. Model training

The prediction accuracy of the multimodal data-driven mechanical performance prediction model is closely related to the sample dataset. In this study, data augmentation technology was used to process EBSD images, and a total of 864 sets of microstructure image samples were obtained. The image data and rolling process data (including TK, RP, AT, DR, CDR, RS, RF, IT, OT) were matched one-to-one to construct numerical and image-based multimodal modeling datasets. To ensure that there are enough training sets for model training and, at the same time, to keep a part of the dataset for validation, the ratio of the divided training set and test set is 4:1. In this study, six models are trained; 20% of the total dataset is used as the test dataset, and the other 80% is used as the training dataset during each training. The mean absolute percentage error (MAPE) is chosen as the loss function in this study because it is widely used in regression problems:

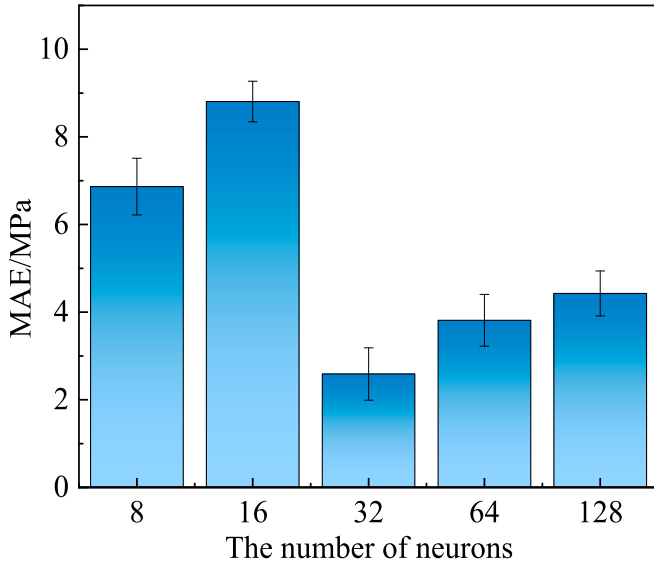


Fig. 11. MAEs of the model for the FC4 layers with different numbers of neurons.

$$\text{loss} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (8)$$

To evaluate the prediction accuracy and performance of the proposed multimodal fusion model, the mean square error (MSE), the root mean square error (RMSE), the average absolute error (MAE) and the coefficient of determination (R^2) are calculated via the following equations:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (9)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\bar{y}_i - y_i)^2} \quad (12)$$

where n is the number of samples, y is the experimentally measured target result, $\hat{y}$ is the prediction result of the corresponding multimodal fusion model, and $\bar{y}$ denotes the average value of y .

The model is trained with a batch size of 8, a learning rate of 0.001, an Adam optimizer, and 200 epochs. The program is written via Keras deep learning software. The operating system is Windows 11, and the hardware is an Intel Core i5-12500 CPU.

4. Results and discussion

4.1. Comparative analysis of model prediction results

To verify the effectiveness of the multimodal model that combines microstructure information for predicting the mechanical properties of stainless steel, this paper proposes six models for comparison: a MLP that uses only rolling process information as input, a CNN that uses only one microstructure image as input (CNN-ONE), a CNN that uses four microstructure images spliced together as input (CNN-FOUR), a residual network that uses four microstructure images spliced together as input (RES-FOUR), a residual network that uses four microstructure images spliced together as input and a CBAM added (RES-FOUR-CBAM), and a multimodal model (MULTI-MODEL) coupled with a multilayer perceptron and a residual network with a CBAM added, using both rolling process information and four microstructure images as inputs. It should be noted that the structural parameters of each neural network model are set to the same level as those in the multimodal model branch. The error calculation results of each prediction model are shown in Table 5.

A histogram of the error distribution plotted for each model error result is shown in Fig. 11. As shown in Table 5 and Fig. 12, the MULTI-MODEL, which integrates the microstructure information and information of the rolling process of the SUS316 ultrathin strip is significantly better than the model that uses only the information of a single modality, which also indicates that the information between different modalities is complementary. The accuracy of mechanical property prediction models can be significantly improved by using both microstructure information and rolling process information.

To evaluate the impact of the number of images, the network structure, and the CBAM module on the prediction accuracy, only the microstructure feature extraction network is investigated in this section.

Fig. 12(c) and (e) show the variation in the MAPE when the numbers of images per sample are one and four, respectively; the loss stabilizes after 50 iterations when the one image is input; furthermore, approximately 90% of the MAPE data on the test set are between 3% and 4.5%. When four image are input, the combined loss stabilizes after 25 iterations, which is faster than inputting a single image, and the relative frequency of the MAPE on the test set is also significantly concentrated between 2.5% and 4%, which is better than the results for a single input image. As shown in Fig. 13(d) and (f), the scattering of points around the $y=x$ line is better in Fig. 13(f) than in Fig. 13(d). This finding indicates that a single EBSD image reflects limited microstructural features and that the joint prediction with multiple images can effectively improve the prediction accuracy of the model.

The model structure is also investigated in this work, using an ordinary CNN and ResNet with a residual connectivity structure, and on the basis of the above study, each sample is input with the stitching of four images. As shown in Fig. 13 (e)(f) and (g)(h), the ResNet based on the residual connection structure is better than the CNN using the same dataset in terms of both MAPE loss and prediction accuracy. The ResNet based on the residual connection structure effectively solves the problem of the CNN of gradient vanishing and improves the prediction accuracy of the model.

To verify the effectiveness of the CBAM module, this study compares

Table 5
Comparative analysis of the forecasted results.

Method	Model Type	R^2		MSE		MAPE	
		train	test	train	test	train	test
Rolling Process data	MLP(A)	0.942	0.917	696.405	861.445	4.247	4.577
Microstructure data	CNN-ONE (B1)	0.929	0.902	921.639	996.307	3.299	3.150
	CNN-FOUR (B2)	0.934	0.912	808.504	922.741	3.191	3.428
	RES-FOUR (B3)	0.946	0.923	668.726	888.392	2.715	3.219
	RES-FOUR-CBAM (B4)	0.955	0.929	652.147	811.239	1.984	2.239
Multimodal data	MULTI-MODEL (C)	0.999	0.998	17.401	19.715	0.683	0.727

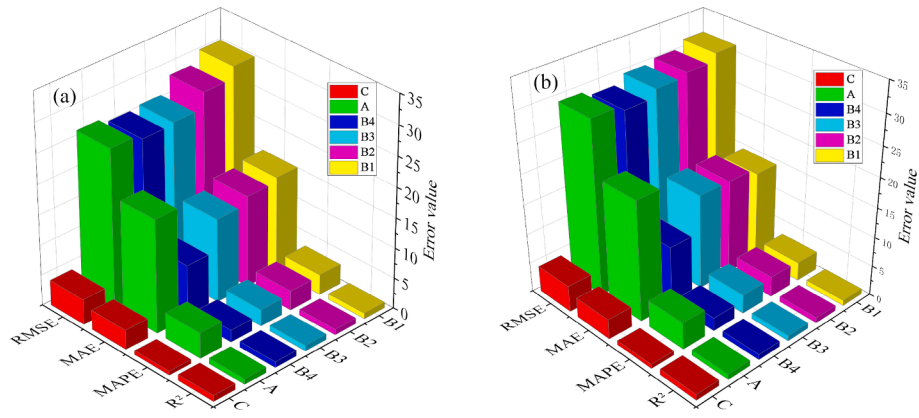


Fig. 12. Histogram of the error distributions for the six models. (a) training set; (b) test set.

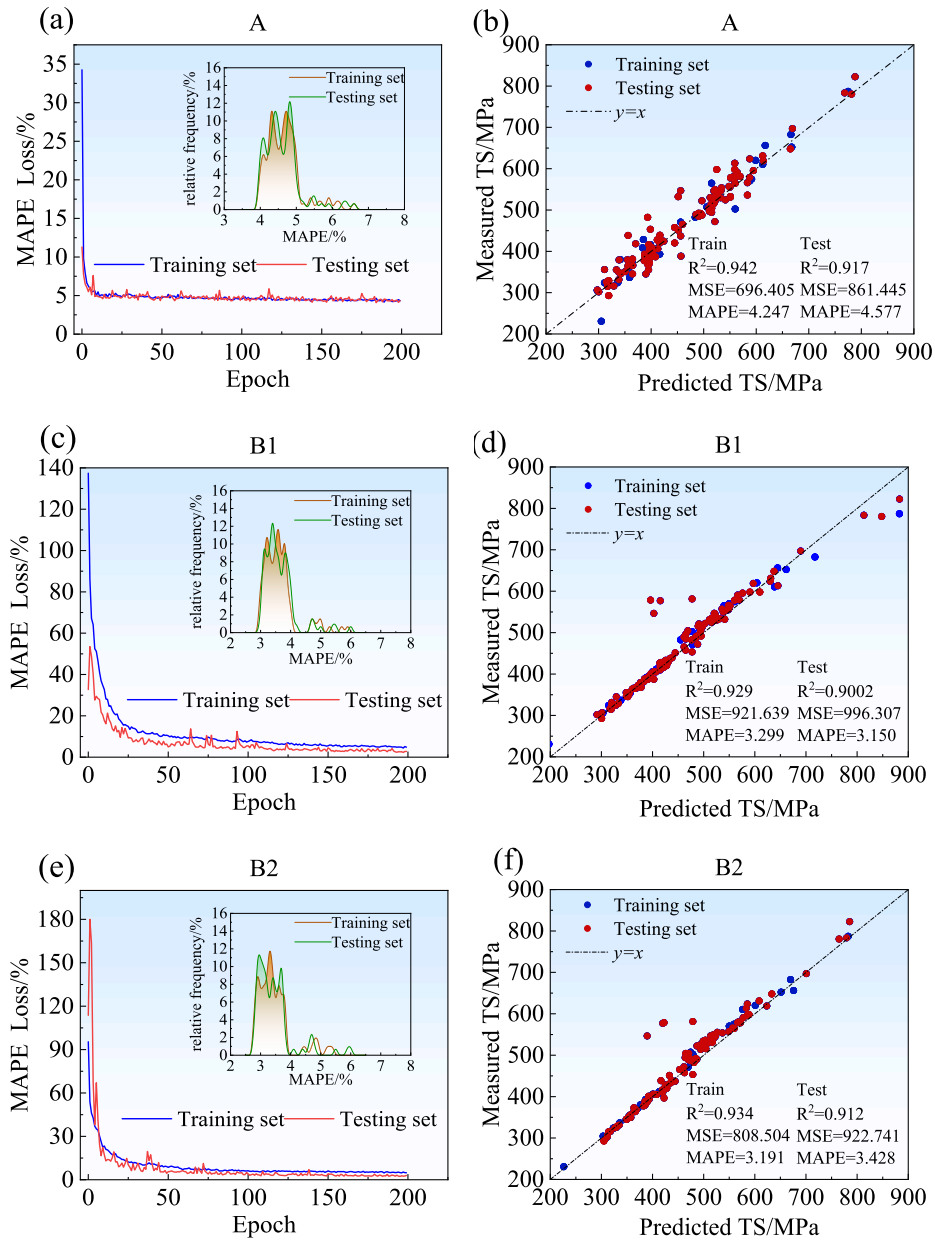


Fig. 13. Training and testing losses and relative frequencies of loss functions during the training of each model and comparison of experimental and predicted results for tensile strength.

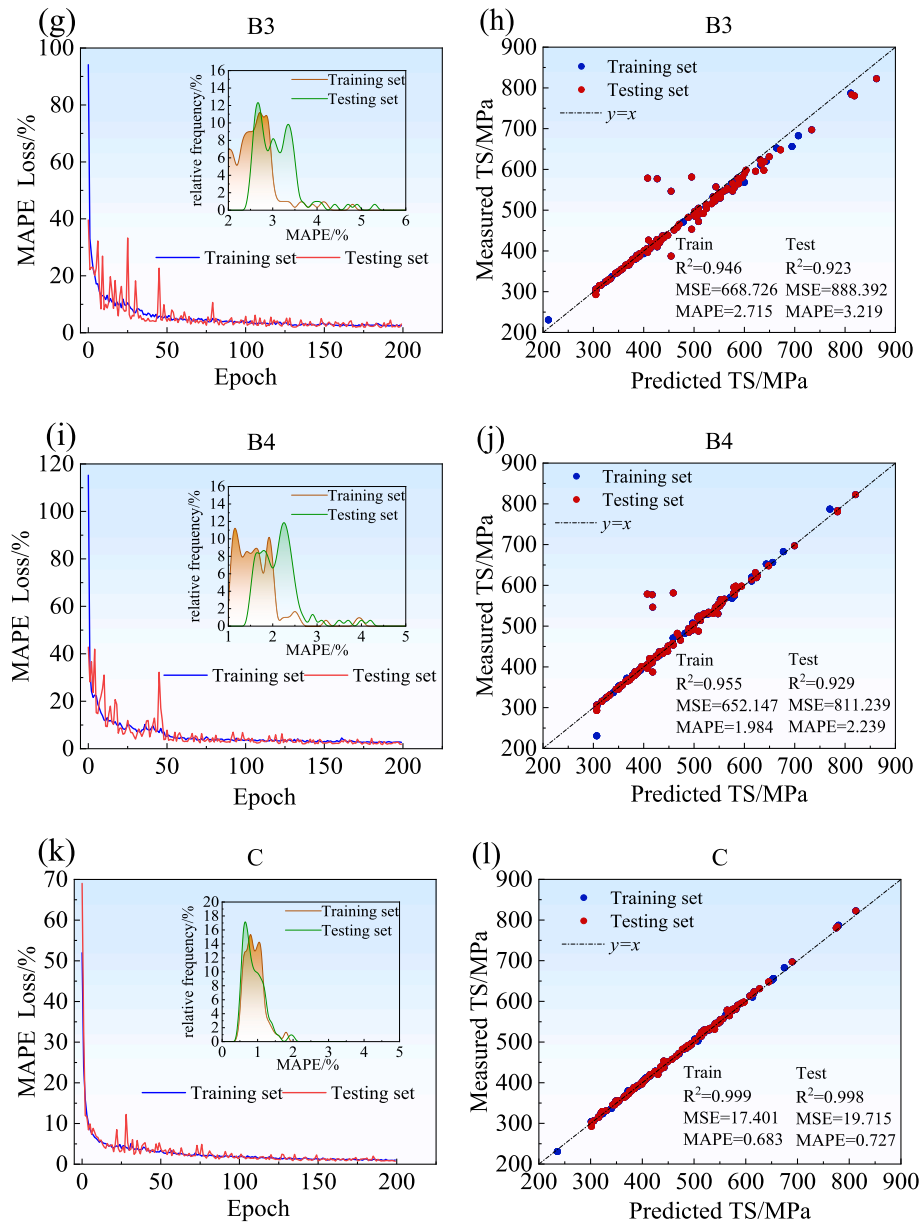


Fig. 13. (continued).

the prediction results of the ResNet with the addition of the CBAM and the ResNet without the addition of the CBAM, as shown in Fig. 13(h) and (j). There is not a large difference in the fitting effect of the two models, but according to Fig. 13(g) and (i), the MAPEs of both the training set and the test set for the ResNet with the added CBAM are lower than those of the ResNet without the added CBAM, and according to their relative frequencies, the models are stable, which verifies the effectiveness of the CBAM module.

4.2. Multimodal fusion model prediction

The above studies show the optimality of the ResNet with the addition of the CBAM on the basis of four microstructure image datasets, which is discussed in this section for the multimodal fusion model built by coupling this network with the MLP.

As shown in Fig. 13(l), the R^2 values of MULTI-MODEL, which integrates the rolling process data and the microstructure image data, are 0.999 and 0.998 for the training set and the test set, respectively, as expected, which indicates that MULTI-MODEL has excellent prediction

accuracy and generalizability and that the model does not lead to obvious overfitting, as observed from Fig. 13(k).

Notably, the R^2 values of the MLP model using only rolling process data are 0.942 and 0.917 for the training set and test machine, respectively. As shown in Fig. 13(a), the loss of the MLP model tends to be stabilize after 20 iterations, and the rate of convergence is the fastest among those of the models demonstrated in this study; however, its prediction accuracy is poor. Compared with multimodal fusion prediction models, microstructure feature extraction networks that use only EBSD images also exhibit poor accuracy and generalization capabilities. This suggests that a single rolling process data type or microstructure image data do not provide sufficient information to predict the mechanical properties of a sample. Therefore, the fusion of multimodal data is a very promising approach for expanding information capacity due to the inherent shortcomings of a single data type.

In addition, as shown in Fig. 13(k), MULTI-MODEL predicts the tensile strength of ultrathin strips of stainless steel with almost 90% of the MAPE data being between 0% and 1.5% on both the training and test sets. According to the prediction results shown in Fig. 13(l), the data

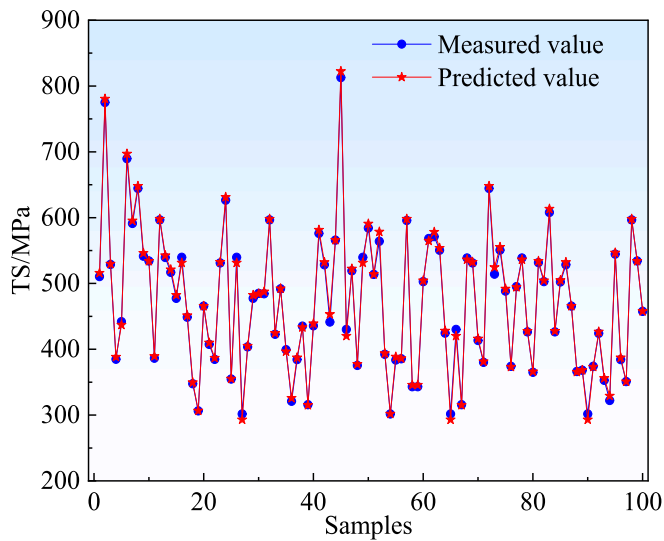


Fig. 14. Comparative analysis of tensile strength.

points are centrally distributed around the straight $y=x$ line, which indicates that most of the predictions are in good agreement with the experimental results. Moreover, the MSE is low enough for both the training and test sets, and R^2 is very close to 1.

Compared with the size of image data (60 KB), that of rolling process data (4B) usually seem insignificant. Therefore, potential features carried by the rolling process data may be overlooked if multimodal data are integrated directly at the input layer. Multimodal fusion modeling, as an adaptive approach, can effectively eliminate this deficiency. Although only microstructure images and rolling process data are used in the current application, the proposed multimodal fusion approach can be easily extended to handle more data types by adding additional branches.

In the comparative study, a thorough analysis is conducted on the stability of the model. The six models constructed are repeatedly trained 100 times, and the MAPE errors on the training and test sets are statistically analyzed after each training. The frequency of MAPE error segmentation intervals in 100 training results is counted, and based on this, the frequency distribution of MAPE error during multiple repeated training processes is plotted, as shown in Fig. 13 (a) (c) (e) (g) (i) (k). From the Fig. 13, it can be seen that the MAPE errors of various models have small differences in the results after multiple training sessions. The error values with high frequency are concentrated in a specific interval. Specifically, after multiple training sessions, the MAPE error of the MLP model is mainly distributed in the range of 4%~5%. The MAPE errors of CNN-ONE, CNN-ONE, CNN-FOUR, RES-FOUR, and RES-FOUR-CBAM models are almost distributed within the range of 3%~4%, 2.5%~4%,

2%~4% and 1%~3%, respectively. The MULTI-MODEL has the highest accuracy and the most stable performance on both the training and testing sets, with MAPE error distribution ranging from 0.5% to 2%.

4.3. Validation of the multimodal model

The 100 samples subjected to different rolling processes in the test set of the multimodal fusion model for predicting the mechanical properties of stainless steel ultrathin strips are selected, the predicted and tested tensile strengths are compared, and the comparison results are shown in Fig. 14. Fig. 14 shows that the tensile strength error of each sample is very low and that the tensile strengths basically overlap.

To demonstrate that the modeling framework is equally applicable to constructing yield strength and elongation prediction models, the established model is used to predict the yield strength and elongation of SUS316 ultrathin stainless steel strips, and 100 samples subjected to different rolling processes are also selected to compare the predicted and tested values of yield strength and elongation; the comparison results are shown in Fig. 15. Fig. 15 shows that the model also shows good prediction accuracy in response to other classes of mechanical properties. As shown in Fig. 16 (a) and (c), the training loss MAE of the multimodal model is concentrated between 1% and 3% for both yield strength and elongation as the target variables of the model, which demonstrates the good stability of the model. As shown in Fig. 16 (b) and (d), the multimodal model also exhibits outstanding fitting results in predicting yield strength and elongation. In summary, the multimodal model framework proposed in this study can be used to predict various mechanical performance parameters and has a wide range of applicability.

5. Conclusion

In conclusion, this work proposes a multimodal deep learning framework to predict the mechanical properties of ultrathin stainless steel strips, and a rolling process-microstructure-mechanical property relationship was successfully established for several different rolling processes and annealing temperatures.

- The information contained in the microstructure images contributes significantly to the prediction of mechanical properties. The correlation coefficient R^2 is 0.902 on the test set for the CNN model with only one input image, whereas the correlation coefficient R^2 is 0.912 on the test set for the CNN model with four input image, which reflects the importance of the amount of information in the dataset used to train the model.
- The ResNet model, which is based on the residual connection structure, has better prediction accuracy than does the traditional CNN model and solves the gradient disappearance problem during the training of the traditional CNN model. Moreover, the

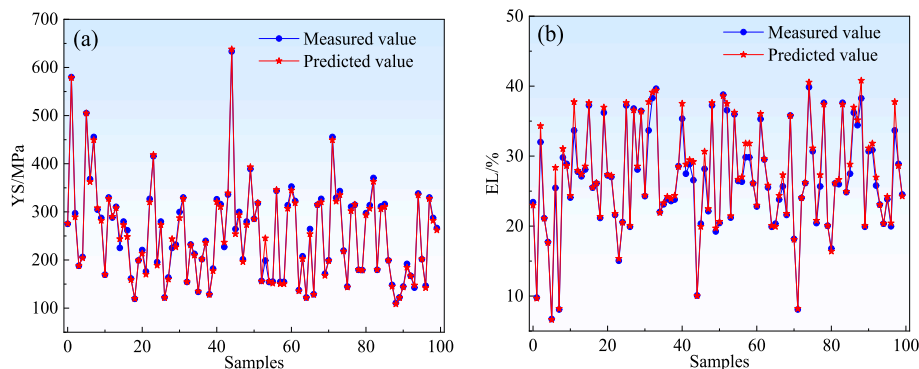


Fig. 15. (a) Comparative analysis of yield strength (b) Comparative analysis of elongation.

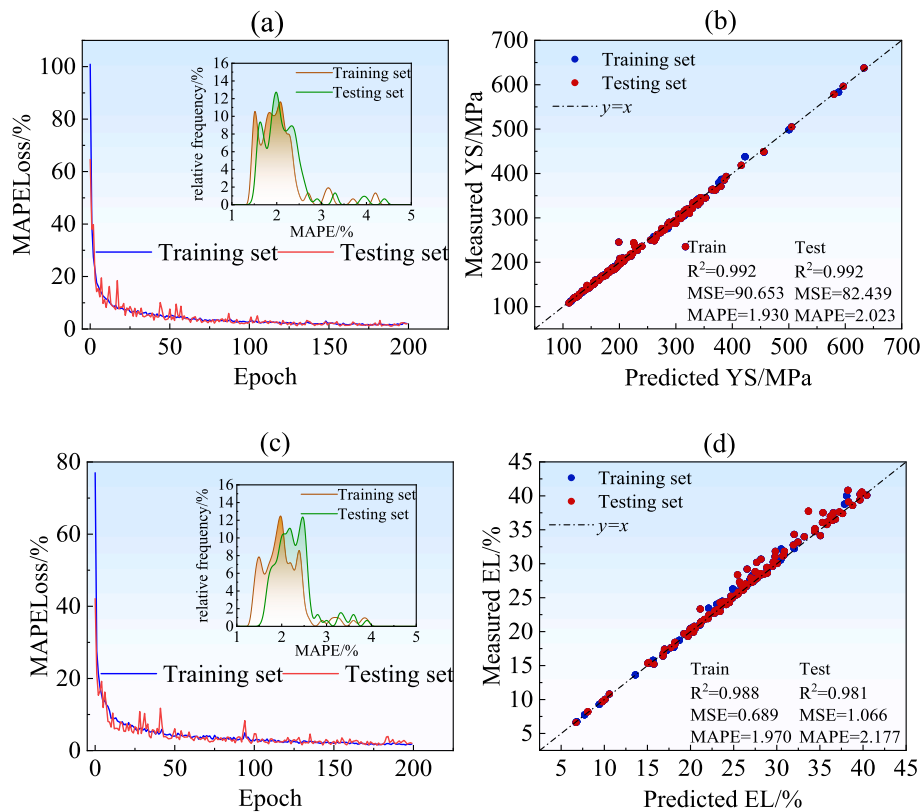


Fig. 16. Training and testing losses and relative frequencies of loss functions during the training of each model and comparison of experimental and predicted results for yield strength and elongation. (a), (b) yield strength, (c) (d) elongation.

addition of the CBAM allows the model to pay more attention to the important parts of the feature map and micro organized images that have an impact on the final task, improves the model's ability to extract features of micro organized images, increases the model's iterative speed, and further improves the prediction accuracy.

- C) The coupled input of rolling process data and microstructure images achieves accurate prediction of the mechanical properties of ultrathin stainless steel strips. The proposed multimodal data fusion model for prediction mechanical properties outperforms the traditional machine learning strategy of analyzing various forms of data independently and forms a multimodal data fusion mining system. The $R^2=0.998$, MSE=19.715, and MAPE=0.727 on the test set, which demonstrates good prediction accuracy. The model also demonstrated good prediction accuracy in predicting yield strength and elongation, indicating its strong applicability.

The predictive model established in this work can be used for rolling process optimization, that is, based on the target mechanical properties, the optimal combination of production process parameters can be inferred, providing decision support for the optimization of production processes. In addition, models can serve as important auxiliary tools in the development of new stainless steel strip production processes. By simulating and predicting the new process conditions, evaluating the impact of the new process on mechanical properties, and screening potential process solutions in advance, the number and cost of actual experiments can be reduced.

Future research work can be continuously promoted from the following aspects.

1 Further expand the modeling samples, use data augmentation techniques to generate more metal microstructure images, and improve the generalization ability and prediction accuracy of the model through

richer sample training. 2. Applying the latest deep learning algorithms to the model proposed in this article to improve the convergence speed of the model while improving the training results. 3. By combining the model presented in this article with transfer learning techniques and making appropriate adjustments and optimizations, the model can be extended to other metal plate and strip rolling process datasets, such as aluminum alloys, copper alloys, etc., to achieve accurate prediction of mechanical properties of more metal materials.

CRediT authorship contribution statement

Zhenhua Wang: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition. **Pengzhan Wang:** Writing – original draft, Validation, Software. **Yunfei Liu:** Investigation, Formal analysis. **Yuanming Liu:** Writing – review & editing, Funding acquisition. **Tao Wang:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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